

Lec: 11

GRAM-SCHMIDT procedure for orthogonalization

Given basis for subspace

$$\overline{x}_1, \overline{x}_2, \dots, \overline{x}_n$$



$$\overline{v}_1, \overline{v}_2, \dots, \overline{v}_n$$

Basis for same subspace
such that

$$\|\overline{v}_i\| = 1$$

$$\& \quad \overline{v}_i \perp \overline{v}_j = 0.$$

$$\overline{v}_1, \overline{v}_2, \dots, \overline{v}_n$$

orthonormal set of vectors

'Orthonormal basis' for the
same subspace.

$\Rightarrow$ any vector in original subspace
can be expressed as linear
combination of $\overline{v}_1, \overline{v}_2, \dots, \overline{v}_n$

Procedure: $\overline{x}_1, \overline{x}_2, \dots, \overline{x}_n$

$$\overline{v_1} = \frac{\overline{x_1}}{\|\overline{x_1}\|}$$

unit norm

$$\begin{aligned}\tilde{v}_2 &= \overline{x_2} - \underbrace{\langle \overline{x_2}, \overline{v_1} \rangle}_{\text{component along } \overline{v_1}} \cdot \overline{v_1} \\ &= \overline{x_2} - (\overline{x_2}^H \overline{v_1}) \overline{v_1}\end{aligned}$$

projection along $\overline{v_1}$.

$\tilde{v}_2$ is $\perp$ to $\overline{v_1}$.

$\tilde{v}_2$ is orthogonal to $\overline{v_1}$.

$$\overline{v_2} = \frac{\tilde{v}_2}{\|\tilde{v}_2\|}$$

unit norm

+ orthogonal to $\overline{v_1}$

$$\tilde{v}_3 = \overline{x_3} - \langle \overline{x_3}, \overline{v_1} \rangle \overline{v_1} - \langle \overline{x_3}, \overline{v_2} \rangle \overline{v_2}$$

$$\overline{v_3} = \frac{\tilde{v}_3}{\|\tilde{v}_3\|}$$

Similarly,

$$\tilde{v}_i = \overline{x_i} - \sum_{j=1}^{i-1} (\overline{x_i}^H \overline{v_j}) \overline{v_j}$$

Removing components along

$$\bar{v}_1, \bar{v}_2, \dots, \bar{v}_{i-1}$$

$$\bar{v}_i = \frac{\tilde{v}_i}{\|\tilde{v}_i\|}$$

This eventually leads to the orthonormal basis $\bar{v}_1, \bar{v}_2, \dots, \bar{v}_n$.

Example:

$$\bar{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \bar{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \quad \bar{x}_3 = \begin{bmatrix} 1 \\ -2 \\ 3 \\ -1 \end{bmatrix}$$

$\|\bar{x}_1\| = \sqrt{4} = 2$

Gram-Schmidt procedure

$$\bar{v}_1 = \frac{\bar{x}_1}{\|\bar{x}_1\|} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{v}_2 = \bar{x}_2 - (\bar{x}_2^T \bar{v}_1) \bar{v}_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - [1 \ 2 \ 3 \ 4] \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \left[\frac{1}{2} + 1 + \frac{3}{2} + 2 \right] \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} - \frac{5}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \end{bmatrix} \longrightarrow ||\tilde{v}_2|| = \sqrt{\frac{9}{4} + \frac{1}{4} + \frac{1}{4} + \frac{9}{4}} = \sqrt{5}$$

$$\bar{v}_2 = \frac{\tilde{v}_2}{||\tilde{v}_2||} = \frac{1}{2\sqrt{5}} \begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

$$||\bar{v}_2||^2 = \frac{1}{20} [9 + 1 + 1 + 9] = 1$$

$$\Rightarrow ||\bar{v}_2|| = 1$$

$$c_4 \quad \bar{v}_2^T \bar{v}_1 = \frac{1}{4\sqrt{5}} [-3 + 1 + 1 + 3] = 0.$$

$$\tilde{v}_3 = \bar{x}_3 - \langle \bar{x}_3, \bar{v}_1 \rangle \bar{v}_1 - \langle \bar{x}_3, \bar{v}_2 \rangle \bar{v}_2$$

$$= \begin{bmatrix} 1 \\ -2 \\ 3 \\ -1 \end{bmatrix} - \frac{1}{2} \cdot \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \left(\frac{-1}{2\sqrt{5}} \right) \left(\frac{1}{2\sqrt{5}} \right) \begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -2 \\ 3 \\ -1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{1}{20} \begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

$$\tilde{v}_3 = \frac{1}{20} \begin{bmatrix} 12 \\ -46 \\ 56 \\ -22 \end{bmatrix}$$

$$\overline{v}_3 = \frac{\tilde{v}_3}{\|\tilde{v}_3\|} = \frac{1}{\sqrt{5880}} \begin{bmatrix} 12 \\ -46 \\ 56 \\ -22 \end{bmatrix}$$

$$= \frac{1}{14\sqrt{30}} \begin{bmatrix} 12 \\ -46 \\ 56 \\ -22 \end{bmatrix}$$

Orthonormal Basis

$$\frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \frac{1}{2\sqrt{5}} \begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \end{bmatrix}, \quad \frac{1}{14\sqrt{30}} \begin{bmatrix} 12 \\ -46 \\ 56 \\ -22 \end{bmatrix}$$

